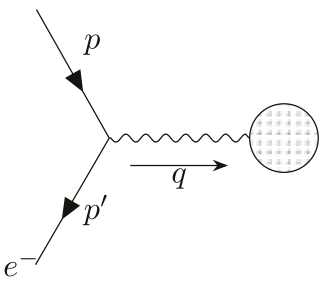
6.2 Equivalent Photon Approximation

Source: Peskin & Schroeder, An Introduction to Quantum Field Theory, Problem 6.2.

Consider the process in which electrons of very high energy scatter from a target. In leading order in α, the electron is connected to the target by one photon propagator:

$i\mathcal{M} =$



(1.1)

where the blob is, in general, complicated coupling of the virtual photon to the target.

If the initial and final energies of the electron are E and E'. Under the high-energy limit,

$$p = (E, 0, 0, E), \quad p' = (E', E' \sin\theta, 0, E' \cos\theta), \tag{1.2}$$

which implies

$$q^2 = (p' - p)^2 = -2EE'(1 - \cos\theta). \tag{1.3}$$

In the limit of forward scattering, whatever the energy loss, the photon momentum approaches $q^2 = 0$; thus the reaction is highly peaked in the forward direction. It is tempting to guess that, in this limit, the virtual photon becomes a real photon. Let us investigate in what sense that is true.

(a) The matrix element of the above process is given by

$$\mathcal{M} = (-ie)\bar{u}(p')\gamma^\mu u(p) \frac{-ig_{\mu\nu}}{q^2} \hat{\mathcal{M}}^\nu(q), \tag{1.4}$$

where $\hat{\mathcal{M}}^\nu(q)$ represents the (in general, complicated) coupling of the virtual photon to the target. Let us analyse the structure of the piece $\bar{u}(p')\gamma^\mu u(p)$. Let $q = (q^0, \vec{q})$ and $\tilde{q} = (q^0, -\vec{q})$. We can expand the spinor products as

$$\bar{u}(p')\gamma^\mu u(p) = Aq^\mu + B\tilde{q}^\mu + C\epsilon_1^\mu + D\epsilon_2^\mu, \tag{1.5}$$

where A, B, C, D are functions of the scattering angle and energy loss and ϵ_1, ϵ_2 are two unit vectors transverse to $\vec{q}$. By dotting the expression with q^μ, we find

$$q_\mu \cdot [\bar{u}(p')\gamma^\mu u(p)] = Aq^2 + Bq \cdot \tilde{q}. \tag{1.6}$$

By Ward identity, we have $q_\mu \bar{u}(p')\gamma^\mu u(p)] = 0$. Thus, we have

$$B = -A\frac{q^2}{q\cdot\tilde{q}} \approx A\frac{2EE'(1-\cos\theta)}{(q^0)^2+\vec{q}^2} \sim \theta^2. \tag{1.7}$$

Thus, B can be ignored in the case of forward scattering.

Again, by Ward identity, we have $q\cdot\hat{\mathcal{M}}(q) = 0$. Thus, A can be ignored as well.

(b) In the frame where $p = (E,0,0,E)$, the spinor $u_s(p)$ is given by

$$u_s(p) = \begin{pmatrix} \sqrt{p\cdot\sigma}\xi_s \\ \sqrt{p\cdot\bar{\sigma}}\xi_s, \end{pmatrix} \tag{1.8}$$

where

$$\sigma^3\xi_\pm = \pm\xi_\pm \implies \xi_+ = \begin{pmatrix}1\\0\end{pmatrix}, \quad \xi_- = \begin{pmatrix}0\\1\end{pmatrix}, \tag{1.9}$$

and

$$\sqrt{p\cdot\sigma} = \sqrt{E(1-\sigma^3)}, \quad \sqrt{p\cdot\sigma} = \sqrt{E(1+\sigma^3)}. \tag{1.10}$$

Thus,

$$\boxed{u_+(p) = \sqrt{2E}\begin{pmatrix}0\\0\\1\\0\end{pmatrix}, \quad u_-(p) = \begin{pmatrix}0\\1\\0\\0\end{pmatrix}.} \tag{1.11}$$

Next, let us evaluate $u(p')$. Using $p' = (E', E'\sin\theta, 0, E'\cos\theta)$, we find

$$\hat{p}'\cdot\vec{\sigma}\xi_\pm = \pm\xi_\pm \implies \xi_+ = \begin{pmatrix}\cos\frac{\theta}{2}\\ \sin\frac{\theta}{2}\end{pmatrix}, \quad \xi_- = \sqrt{2E}\begin{pmatrix}-\sin\frac{\theta}{2}\\ \cos\frac{\theta}{2}\end{pmatrix} \tag{1.12}$$

and

$$\sqrt{p\cdot\sigma} = \sqrt{E'(1-\hat{p}\cdot\vec{\sigma})}, \quad \sqrt{p\cdot\bar{\sigma}} = \sqrt{E'(1+\hat{p}\cdot\vec{\sigma})}. \tag{1.13}$$

Thus, we find

$$\boxed{u_+(p') = \sqrt{2E'}\begin{pmatrix}0\\0\\\cos\frac{\theta}{2}\\\sin\frac{\theta}{2}\end{pmatrix}, \quad u_-(p') = \sqrt{2E'}\begin{pmatrix}-\sin\frac{\theta}{2}\\\cos\frac{\theta}{2}\\0\\0\end{pmatrix}.} \tag{1.14}$$

The polarisation vectors are defined to be parallel and horizontal to the scattering plane ($x - z$ plane). Thus,

$$\boxed{\epsilon_1^\mu = \frac{1}{\sqrt{E^2 + E'^2 - 2EE'\cos\theta}}(0, E'\cos\theta - E, 0, -E'\sin\theta), \quad \epsilon_2^\mu = (0,0,1,0).} \tag{1.15}$$

Now, let us evaluate $\bar{u}(p')\vec{\gamma}\cdot\vec{\epsilon}_i u(p)$. With some algebra, one should find

$$\boxed{\begin{aligned}&\bar{u}_\pm(p')\vec{\gamma}\cdot\vec{\epsilon}_1 u_\pm(p) = -\frac{E'+E}{|E'-E|}\theta\sqrt{EE'} = C,\\ &\bar{u}_\pm(p')\vec{\gamma}\cdot\vec{\epsilon}_2 u_\pm(p) = \pm\theta\sqrt{EE'} = D_\pm,\\ &\bar{u}_\pm(p')\vec{\gamma}\cdot\vec{\epsilon}_1 u_\mp(p) = \bar{u}_\pm(p')\vec{\gamma}\cdot\vec{\epsilon}_2 u_\mp(p) = 0.\end{aligned}} \tag{1.16}$$